F12 4th CAR

Lecture 8

Section 1.3 Linear Equations in Two Variables

Slope as a rate of change

Equation of a verical line

Equation of a horizontal line

Point-slope for of the equation of a line

Slope-intercept form of the equation of a line

Parallel and perpendicular lines

Section 1.4 Functions

Relation

Function

Difference quotient

Piecewise-defined function

Domain of a function

13pgs + 1 prac probs remote p 11 Fig 2 4th CAR 100

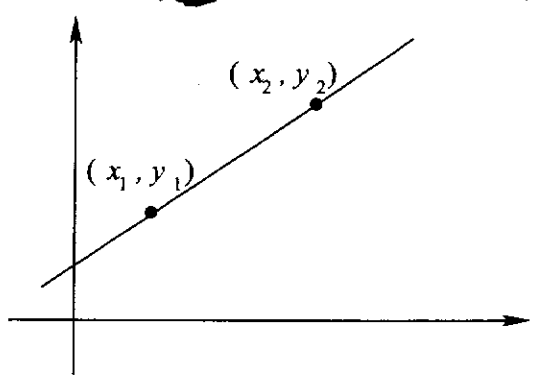
Lecture 8, part I: Section 1.3 Linear Equations in Two Variables

exponents are 1.

Def. The equation $Ax + By + C = 0$ is the general form of a linear equation in two variables.

The **slope** m of L is defined by the formula

$$m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{y_1 - y_2}{x_1 - x_2}$$



Slope as a Rate of Change

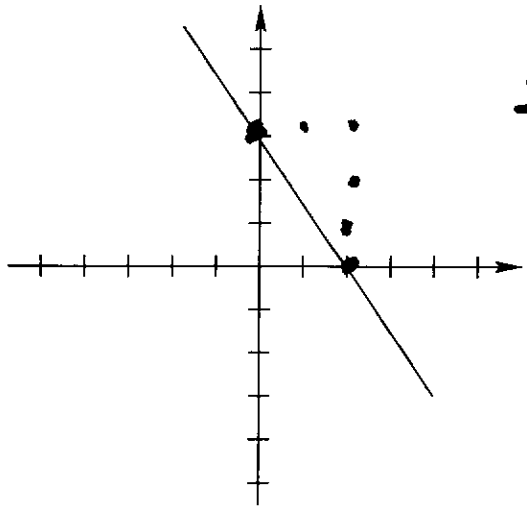
Δ

For line L, m is the quotient of the **change in** y, the **Rise**, and the **change in** x, the **Run**.

We have $m = \dfrac{\text{Rise}}{\text{Run}} = \dfrac{\Delta y}{\Delta x} = \dfrac{y_2 - y_1}{x_2 - x_1} = \dfrac{\text{difference of ys}}{\text{difference of xs}}$

NOTE: The slope of a line measures the **average rate of change of y with respect to x**

ex. Find the slope of the given line below:



$$\frac{\text{rise}}{\text{run}} = \frac{+3}{-2} = -\frac{3}{2} = \frac{-3}{2} = \frac{3}{-2}$$

ex. Find the slope of the line through the points $(-6, 2)$ and $(-5, -4)$

$$m = \frac{2 - {}^{-}4}{-6 - {}^{-}5} = \frac{6}{-1} = -6$$

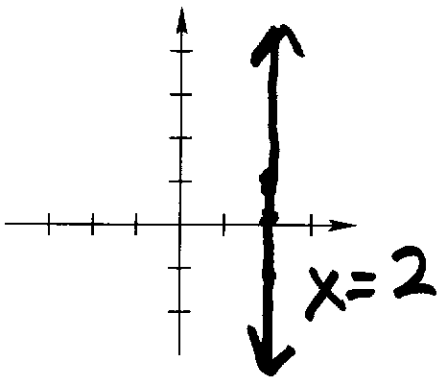
$$m = \frac{-4 - 2}{-5 - {}^{-}6} = \frac{-6}{1} = -6$$

NOTE: The order of subtraction is important!

x	y
2	-1
2	0
2	1

ex. Graph the equation $x = 2$.

y is all real nummbers



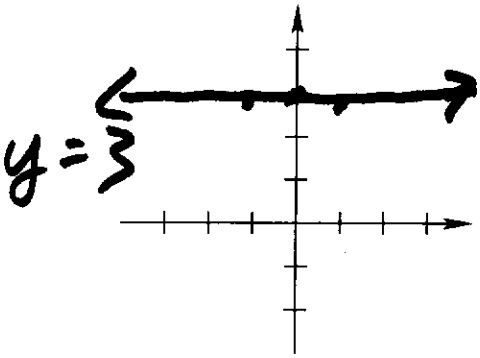
Equation of a Vertical Line

A vertical line with x-intercept a is given by the equation $x = a$

ex. Graph the equation $y = 3$.

x	y
-1	3
0	3
1	3

x is all real numbers

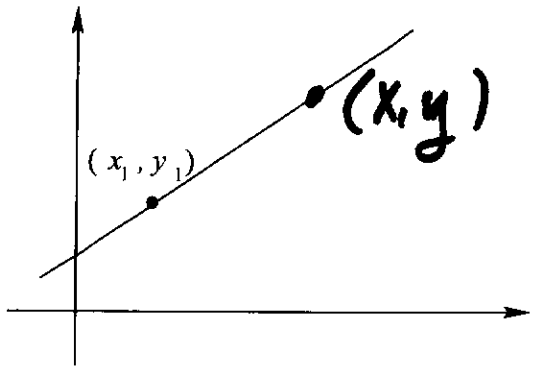


Equation of a Horizontal Line

A horizontal line with y-intercept b is given by the equation $y = b$

$\cancel{m = \frac{y - y_1}{x - x_1}}$ cross mult.

Now consider a nonvertical line L with slope m containing the point (x_1, y_1)



Let (x, y) be any other point on L. we have

Point-Slope Form of the Equation of a Line

A line L with slope m which passes through the point (x_1, y_1) has equation $y - y_1 = m(x - x_1)$

ex. Write the equation of the line with slope 2 and containing $(-1, 1)$.

$y - 1 = 2(x - -1)$

$y - 1 = 2(x + 1)$

$y = 2x + 3$ slope-intercept form of the equation of a line.

general form $\rightarrow$ $2x - y + 3 = 0$

ex. Write ~~the~~ an equation of the line through $(-3, 1)$ and $(3, -2)$.

① Find slope. $m = \frac{1 - -2}{-3 - 3}$

$m = \frac{3}{-6} = -\frac{1}{2}$

$y - y_1 = m(x - x_1)$

$y - -2 = -\frac{1}{2}(x - 3)$

$y + 2 = -\frac{1}{2}x + \frac{3}{2}$

$\boxed{y = -\frac{1}{2}x - \frac{1}{2}}$

$2y = -x - 1$

$x + 2y + 1 = 0$

Slope-Intercept Form of the Equation of a Line

A line L with slope m and the y-intercept $(0, b)$ has equation $y = mx + b$

ex. Write the equation of the line through $(-3, 1)$ and y-intercept $(0, 5)$.

① $m = \frac{1 - 5}{-3 - 0} = \frac{-4}{-3} = \frac{4}{3}$

$y = mx + b$

$y = mx + b$

$\boxed{y = \frac{4}{3}x + 5}$

① $y = mx + b$

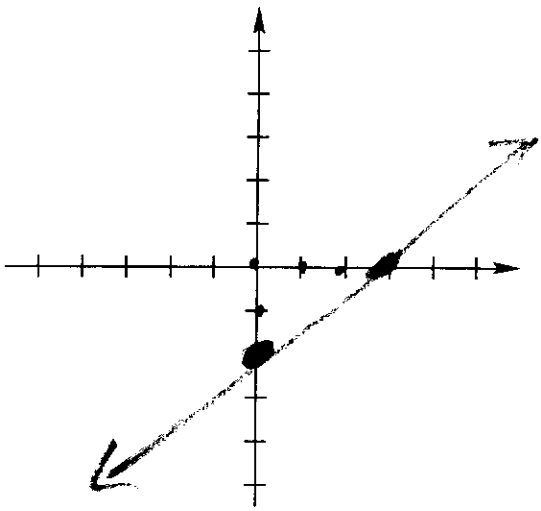
$-3y = -2x + 6$

$y = \frac{2}{3}x - 2$

slope $= \frac{2}{3}$

y-int is -2

ex. Find the slope and y-intercept of the line with equation $2x - 3y = 6$. Graph the equation.



Graphing: ① graph y-int.

② Graph another pt by using

$m = \frac{\text{rise}}{\text{run}} = \frac{2}{3}$

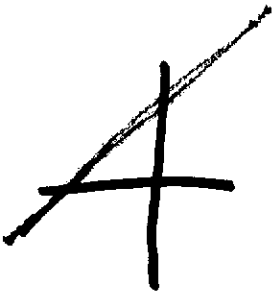
Parallel and Perpendicular Lines

Given distinct nonvertical lines L_1 and L_2 with slopes m_1 and m_2.

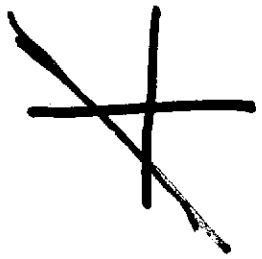
1) L_1 is **parallel** to L_2 if and only if $m_1 = m_2$, y-intercepts are different.

2) L_1 is **perpendicular** to L_2 if and only if the slopes are negative reciprocals of each other.

ex $m = -2$ then $m_\perp = \frac{1}{2}$



pos. slope



neg slope

ex. Find the general form of the equation of the line passing through the point $(1, -3)$ and

1) parallel to the line $2x - 4y - 3 = 0$

① $m = ?$

② $y - y_1 = m(x - x_1)$

$y - {}^-3 = \frac{1}{2}(x - 1)$

$y + 3 = \frac{1}{2}x - \frac{1}{2}$

$y = \frac{1}{2}x - \frac{7}{2}$

$-4y = -2x + 3$

$y = \frac{1}{2}x - \frac{3}{4}$

so $m = \frac{1}{2}$

or $2y = x - 7$

$x - 2y - 7 = 0$

2) perpendicular to the line $2x - 4y - 3 = 0$

① Since $m = \frac{1}{2}$, $m_\perp = -2$

$y - y_1 = m(x - x_1)$

$y - {}^-3 = -2(x - 1)$

$y + 3 = -2x + 2$

$2x + y + 1 = 0$

Applications

ex. When a retailer prices a DVD player at \$90, weekly sales at the district stores average 150 units. He has observed that for each \$5 drop in the selling price, weekly sales increase by an average of 15 units.

1) Find two points of the form (p, x) where x is the number of DVD players that will sell when the price is p dollars.

2) Find the slope of the line through these two points.

3) Interpret the slope in terms of the problem.

Practice.

1) Find the equation of the line containing $(-2, 3)$ and

(a) parallel to the y-axis.

(b) perpendicular to the y-axis.

2) Find k so that the line L through the points $(-2, 8)$ and $(4, k)$ is perpendicular to the line $x - 3y = 1$.

Answer. 1) (a) $x = -2$ (b) $y = 3$ 2) $k = -10$

Lecture 8, part II: Section 1.4 Functions

Def. A **relation** is a correspondence between two sets A and B. A relation is expressed as a set of ordered pairs (x, y), where x is an element of set A and y is the corresponding element of set B.

The **domain** of a relation is the set of all first elements and the **range** is the set of all second elements in the ordered pairs.

Functions represented by ordered pairs

Each x can only have one y.

Def. A **function** from set A into a set B is a relation that assigns to each element x in set A **exactly one** element y in the set B.

ex. Does the given relation represent a function? Find the domain and range of each function.

1) $\{(-1, 0), (2, 4), (0, 0), (2, -2)\}$

$2 \to 4$, $2 \to -2$ not a function

2) $\{(-2, 0), (-1, 2), (0, 0), (1, 2), (2, 0)\}$

Yes, a function

D: $\{-2, -1, 0, 1, 2\}$

R: $\{0, 2\}$

Domain: ① ÷ by 0

② even $\sqrt{\text{neg}}$

$\sqrt[4]{\text{neg}}$ not in reals

NOTE: A function is a relation in which no two ordered pairs have the same first coordinate.

Functions represented by an equation

<u>ex.</u> $y = x^2 - 1$

1) Find the **value** of the function at $x = -1$.

$y = (-1)^2 - 1 = 1 - 1 = \boxed{0}$

2) domain: $(-\infty, \infty)$ or All reals

x-values

3) range:

y-values $\boxed{[-1, \infty)}$

$f(x) = x^2 - 1$

$y = f(x)$ is the **function value** assigned to domain element x (the argument) by f.

To evaluate the function value of a given argument, you replace each x in the function formula with the argument.

ex. If $f(x) = 3 - 2x$, find:

1) $f(-2) = 3 - 2(-2) = 3 + 4 = 7$

2) $f(a) - 1 = 3 - 2a - 1 = -2a + 2$

3) $f(a-1) = 3 - 2(a-1) = 3 - 2a + 2 =$
$= -2a + 5$

NOTE: $f(a) - 1 \neq f(a-1)$

ex. Let $f(x) = x^2 - 2x$, find and simplify the **difference quotient:** $\dfrac{f(x+h) - f(x)}{h}$, $h \neq 0$

$$f(x+h) = (x+h)^2 - 2(x+h) = x^2 + 2xh + h^2 - 2x - 2h$$

$$\frac{f(x+h) - f(x)}{h} = \frac{\cancel{x^2} + 2xh + h^2 - \cancel{2x} - 2h - (\cancel{x^2} - \cancel{2x})}{h}$$

$$= \frac{2xh + h^2 - 2h}{h} = \frac{\cancel{h}(2x + h - 2)}{\cancel{h}} = 2x + h - 2$$

A Piecewise-Defined Function

ex. If $f(x) = \begin{cases} 2x+1, & x \le 0 \\ x^2 - 1, & x > 0 \end{cases}$, find:

1) $f(-1) = 2(-1)+1 = -1$

2) $f(0) = 2(0)+1 = 1$

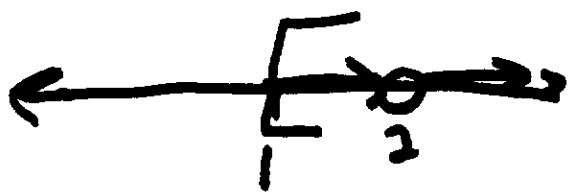
3) $f(2) = 2^2 - 1 = 3$

Domain of a Function

The domain of a function $y = f(x)$ is the set of all real numbers x for which the expression is defined.

ex. Find the domain $f(x) = \dfrac{\sqrt{x-1}}{x^2-4}$

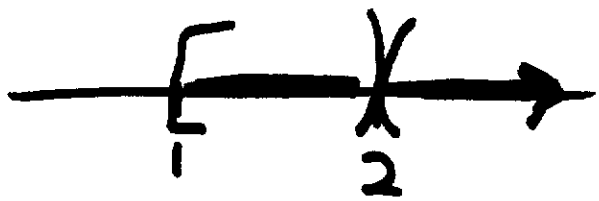
① $x - 1 \ge 0$ $\quad x \ge 1$



and also

② $x^2 - 4 \ne 0$

$x \ne \pm 2$

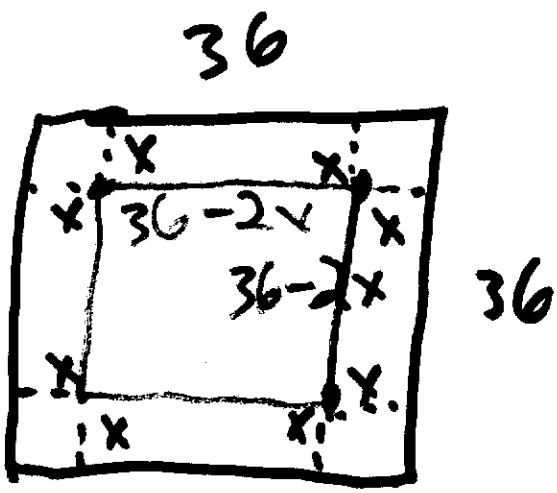


$\boxed{[1, 2) \cup (2, \infty)}$

Applications

<u>ex.</u> An open box is constructed from a square piece of cardboard 36 in. on a side by cutting equal squares (of length x) from each corner and folding up the sides. Express volume V as a function of the length x. Find the domain of the function.

$V = l\,w\,h$



$$\boxed{V = (36-2x)(36-2x)x}$$

D: $0 < 36-2x < 36$

D: $36-2x > 0$

$-2x > -36$

$$\boxed{0 < x < 18 \text{ in}}$$